Assignment Sheet

Task 1 – Direct-Mapped Cache (5 Points)

In a direct mapped cache no replacement policies are necessary, instead the n lowest bits of the memory address are used to determine the line in which the data will be stored. What happens, if instead the n highest bits are used? Does this make the cache more or less efficient in regard to hit rate? Explain why and all concepts you use.

Your answer:

Task 2 – Cache Latency (8 Points)

To ensure that your latest program is cache-friendly, you ran extensive diagnostics and found out that of 8256 cache accesses, 6192 were hits.
The manufacturer of your main memory declares the cache access time as 4ns.
Assuming that this data is a good representation of your machine, and cache accesses are twenty times faster than main memory accesses, what is your cache latency? Give your answer in nanoseconds.

Latency:

Calculations:

Task 3 – Set-Associative Caches (7 Points)

What is the size (in KiB) of a set-associative cache with 8 sets and 4 ways, if the offset is 6 bits long?

Size:

Calculations:

This assignment is due on 05.12.2025 at 10 am.